

Chi-Sheng (Michael) Chen (陳麒升)

ML Researcher & Engineer | Clinical Biosignal ML | LLM Agents & Multi-Agent Systems | Foundation Models

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Nationality: Taiwan | Based in Cambridge, MA, USA

Chi-Sheng (Michael) Chen is a machine-learning researcher and engineer working across clinical biosignal ML and LLM-agent systems. At Harvard/BIDMC he builds multimodal EMS trauma-triage pipelines (ASR → LLM-agent event extraction → structured prediction) over 4M+ patient records; his EEG-based depression-treatment model is in routine outpatient use at Taipei Veterans General Hospital. He has shipped LLM-agent systems (a US-stock research agent suite, a robotics workcell optimizer), cut LLM token usage by 82% for a quant-trading firm, and placed 3rd of 1,300+ teams at Berkeley RDI AgentX-AgentBeats 2026. Prior: Co-Founder/CTO of Neuro Industry (GCP MLOps on 10K+ EEG samples) and Digital IC R&D Engineer at MediaTek. 49 publications, 491 citations (h-index 13); reviewer for NeurIPS, ICML (Gold Reviewer) and 40 journals.

RESEARCH & PROFESSIONAL EXPERIENCE

Harvard Medical School & Beth Israel Deaconess Medical Center

MA, USA

Research Affiliate, Advisor: Prof. Dr. Gabriel Brat

Nov 2024 – Present

- Building a real-time EMS trauma-triage pipeline (ASR → LLM-agent event extraction → structured prediction) over **4M+ patient records** using multimodal AI.
- Collaborating with surgeons on AI-assisted decision support systems.

WhaleForce (subsidiary of VICI Holdings)

Taipei, Taiwan

Fractional Token Efficiency Research Lead (Advisor)

Aug 2026 – Present

- Advising on LLM token-efficiency research and multi-agent architecture for AI-driven quantitative trading; token-efficiency work **cut LLM token usage by 82%**.

Omnis Labs

Remote

Co-Founder

Sep 2024 – May 2026

- AI-driven DeFi liquidity positioning protocol with automated market maker (AMM) strategies and backtesting systems.
- Achieved 50%–120% base fee APR on WBTC/USDC and ETH/USDC pairs (Katana on SushiSwap) using time-series model-driven AMM strategies.

Neuro Industry, Inc.

CA, USA

Researcher, Co-Founder & CTO

Mar 2024 – Jan 2025

- Built GCP-based MLOps pipeline, enabling training on 10K+ EEG samples.
- 3 research papers: quantum machine learning for EEG signal processing (2) and EEG-to-image diffusion (1).

Department of Computer Science, National Yang Ming Chiao Tung University

Hsinchu, Taiwan

Research Assistant, Advisor: Prof. Chun-Shu Wei

Dec 2023 – Aug 2024

- Developed MUSE, a multimodal contrastive learning framework for EEG-based image recognition (32 citations).
- Research on nonstationary EEG dynamics and foundation models for brain-computer interfaces.

MediaTek Inc.

Hsinchu, Taiwan

Digital IC R&D Engineer

Dec 2021 – Oct 2023

- Responsible for microprocessor IP development for flagship 5G smartphones' display and AMBA SoC implement.
- Hardware virtualization architecture RTL design and IP verification with UVM and SystemVerilog.

Department of Surgery, National Taiwan University Hospital

Taipei, Taiwan

Research Assistant, Advisor: Dr. Shuo-Lun Lai

Jul 2021 – Sep 2021

- Deep learning applications on surgery automation with YOLO-based model.
- Developed a prototype for a surgical smoke evacuation system.

Department of Psychiatry, Taipei Veterans General Hospital

Taipei, Taiwan

Research Assistant, Advisor: Prof. Dr. Cheng-Ta Li

Sep 2019 – Jun 2021

- Built ML models on clinical EEG from major-depressive-disorder patients to predict treatment response.

Max Planck Institute for Chemical Physics of Solids (MPI CPfS)

Dresden, Germany

Research Internship, Advisor: Dr. Alexander Komarek & Dr. Li Zhao

Jul 2018 – Sep 2018

- Synthesis and characterization of candidate Co-based chalcogenide superconductors.

Other: OpenAI (contractor via Mercor) — AI Trainer, RLHF preference data & model-behavior evaluation *Mar 2025 – Oct 2025*

AGENT & LLM ENGINEERING

Berkeley RDI AgentX-AgentBeats — 3rd Place / 1,300+ teams (2026). *MadGAA*, a multi-agent OSCE medical evaluation system: role-specialized agents (examiner, patient, judge) + verifiable scoring rubric (RAG + tool use + verifier).

US-stock LLM agent suite (deployed). Browser agent as an explicit PLAN→LOCATE→ACT→VERIFY state machine with fault-injection recovery; layered SEC 10-K extractor; 22 research agents + 11 placebo controls with lookahead-free backtests, per-provider cost ledger, and eval dashboards.

Robotics Workcell Optimizer (live). NL → spec → robot selection → 2D layout → 3D preview via multi-LLM abstraction (Claude/OpenAI/Gemini, validation-repair loop) + OR-Tools CP-SAT; ISO 13855 safety as inviolable hard constraints.

Also: *flyhypo* (connectome + PubMed + LLM → grounded, falsifiable neuron hypotheses); *paper-evidence* (verbatim-quote gate + cross-family judge to block LLM hallucination); two live NYCU RAG physics tutors (Next.js + Gemini + pgvector).

Stack: LangChain/LangGraph, ReAct & function calling, multi-agent orchestration, vLLM, LoRA/QLoRA, SFT/DPO/RLHF, RAG (pgvector/FAISS/Qdrant), OR-Tools, OpenAI/Anthropic/Gemini SDKs.

EDUCATION

National Taiwan University (NTU)

Master of Science in Biomedical Electronics and Bioinformatics
Advisor: Prof. Dr. Cheng-Ta Li, MD & Prof. Chung-Ping Chen

Taipei, Taiwan

Sep 2019 – Jun 2021

- The research EEG model for TMS pattern personalized prediction is currently being applied to real outpatient patients in the Precision Depression Intervention Center (PreDIC), Taipei Veterans General Hospital.

National Chiao Tung University (NCTU)

Bachelor of Engineering in Electrophysics
Bachelor of Science in Interdisciplinary Science Degree Program

Hsinchu, Taiwan

Sep 2015 – Jun 2019

SKILLS

AI/ML: Python, PyTorch, HuggingFace, Scikit-learn, MNE, MLflow, R, MATLAB

Systems: C/C++, JavaScript, Rust, MySQL, Solidity, Git, Docker, GCP, AWS

Hardware: Verilog, SystemVerilog, UVM, HSPICE, Genus, Verdi, SG-Lint, Perl

Languages: Chinese (Native), English (Fluent)

SELECTED PUBLICATIONS

(49 total, 491 cites, h-index 13 — full list on Google Scholar)

[First author unless noted]

Chen, C.S., et al. “FreqLens: Interpretable Frequency Attribution for Time Series Forecasting” *Under review*, ICLR 2027; arXiv:2602.08768.

Chen, C.S., et al. “Large Cognition Model: Towards Pretrained EEG Foundation Model” arXiv preprint (2025), arXiv:2502.17464 — **29 citations**.

Chen, C.S., Wei, C.S. “Mind’s Eye: Image Recognition by EEG via Multimodal Similarity-Keeping Contrastive Learning” arXiv preprint (2024), arXiv:2406.16910 — **32 citations**.

Chen, C.S., et al. “Exploring the Potential of Electroencephalography Signal-Based Image Generation Using Diffusion Models: Integrative Framework Combining Mixed Methods and Multimodal Analysis” JMIR Medical Informatics (2025), <https://doi.org/10.2196/72027> .

Chen, C.S., et al. “Exploring the potential of QEEGNet for cross-task and cross-dataset electroencephalography encoding with quantum machine learning” Journal of Signal Processing Systems 98, 5 (2026), <https://doi.org/10.1007/s11265-025-01979-2> .

Li, C.T., Chen, C.S., et al. “Prediction of antidepressant responses to non-invasive brain stimulation using frontal electroencephalogram signals: Cross-dataset comparisons and validation” Journal of Affective Disorders (2023), <https://doi.org/10.1016/j.jad.2023.08.059> . [2nd author; model deployed at Taipei VGH PreDIC]

INVITED TALKS & PRESENTATIONS

NVIDIA GTC 2026 (Top 8 Finalist, two posters); **MICCAI 2026** — DEMI spotlight talk and CLINICCAI poster; **IEEE ICASSP 2026** (two orals) and **2025** (oral); **IEEE Quantum Week 2025**; **DH2026** (two short papers); **QTML 2024** (poster).
Guest lectures at NYCU ECE (Wearable Chip Design), NTU CSIE (Advanced Compilers) and NYCU Maker Club; invited seminar, Dept. of Applied Physics, Tunghai University (2025–2026).

PROFESSIONAL SERVICE

Conference Reviewer (2026–2027): NeurIPS, ICML (Gold Reviewer — top tier), AAAI, KDD, MICCAI, ICASSP, NLDL 2027, IEEE QAI 2026. **Program Committee:** QNRL Workshop @ IEEE WCCI 2026.

Journal Reviewer (2025–present): 40 journals incl. TPAMI, TNNLS, TASLP, JBHI, Neural Networks, npj Quantum Information, Scientific Reports; **IOP Trusted Reviewer** (2026).

AWARDS & ACHIEVEMENTS

2026 NVIDIA GTC Top 8 Finalist: “GPU-Accelerated Chinese Brain-to-Text Interface Powered by NVIDIA H100 80GB.”

2026 Berkeley RDI AgentX-AgentBeats: Phase 1 Healthcare Agent 3rd Place (Team MadGAA, 1,300+ teams worldwide).

2023 Certificate of the 20th National Innovation Award, clinical research category.

2019 International Blockchain Olympiad (IBCOL): World Finalist competition with 100+ teams.

2017 Microsoft International Imagine Cup : 2nd place in Taiwan.

2016 International Genetically Engineered Machine Competition (iGEM): World Gold medal, Best Applied Design, Best Part Collection, Best Presentation competition with 300+ university teams.